

Week 3: Fundamentals of Probability and Statistics I

PLSC 309 Quantitative Political Analysis

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Reminder

Problem Set 1

Due: tomorrow, September 9 at 12:00 p.m. (noon)

Submit it on Canvas

Final Project: Checkpoint 1

Due: next Wednesday, September 16 at 11:59 p.m.

Instructions: Posted on Canvas

Worth 5 points

What to Submit

No longer than 2 pages, 12 pt, including:

- **Research Question**
- **Theory**
- **One Hypothesis**
- **Possible Data Source(s)**
 - Is there a dataset available that could help you answer your question?

Office Hours Schedule Change

This week

Office hours will meet as usual on

Wednesday, September 9, 1:00–2:30 p.m.

Beginning next week

Office hours will move to

Mondays, 1:00–2:30 p.m.

Location remains the same: 553 Susan Welch Building

What Did We Learn Last Time?

- Operationalization
- Types of Variables
- Types of Data

Today's Agenda

Fundamentals of Probability and Statistics I (1).

- Random Variable
- Probability Distribution
- Expected Value
- Variance and Standard Deviation
- Covariance and Correlation

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#).

Part 1. What is a Random Variable?

The Pollster's Problem

It's October 2024. You're a pollster trying to predict the presidential election.

You call a randomly selected voter. Before they answer, several things are uncertain:

- **Will they answer the phone?** (Yes / No)
- **Who do they support?** (Harris / Trump / Other / Undecided)
- **How certain are they?** (Very / Somewhat / Not at all)
- **Will they actually vote?** (Definitely / Probably / Probably not / Definitely not)

These are all **qualitative (categorical) outcomes**.

But ultimately, you need to produce a **number**: "Harris: ?%, Trump: ?%"

From Outcomes to Numbers

To do statistics, we need to **convert qualitative outcomes into numbers**.

Outcome	Variable	Possible Encoding
Supports Harris	Harris	$Harris = 1$ (Trump/Other/Undecided=0)
Supports Trump	Trump	$Trump = 1$ (Harris/Other/Undecided=0)
Definitely will vote	Vote	$Vote = 4$
Probably will vote	Vote	$Vote = 3$
Probably won't vote	Vote	$Vote = 2$
Definitely won't vote	Vote	$Vote = 1$

The function that performs this translation is called a **random variable**.

Definition: Random Variable

Random Variable

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ that assigns a real number to each outcome in the sample space.

Key insight: A random variable is *neither random nor a variable*. It's a **function**.

- The *input* is an outcome $\omega \in \Omega$ (which outcome occurs)
- The *output* is a number $X(\omega) \in \mathbb{R}$ (the numerical value)

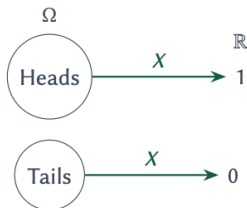
The “**randomness**” comes from not knowing which ω will occur.

Convention: Capital letters (X, Y, Z) for **random variables**; lowercase letters (x, y, z) for **specific values**.

Example: Coin Flip

Sample space: $\Omega = \{\text{Heads}, \text{Tails}\}$

Random variable: $X(\omega) = \begin{cases} 1 & \text{if } \omega = \text{Heads} \\ 0 & \text{if } \omega = \text{Tails} \end{cases}$



Before the flip: X is uncertain (could be 0 or 1). After: X takes a specific value.

Example: Die Roll

Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$ (outcomes are already numbers!)

Natural random variable: $X(\omega) = \omega$ (the identity function)

But we could define *other* random variables on the same sample space:

ω	1	2	3	4	5	6
$X(\omega) = \text{face value}$	1	2	3	4	5	6
$Y(\omega) = \text{is it even?}$	0	1	0	1	0	1
$Z(\omega) = \text{is it } \geq 5?$	0	0	0	0	1	1

Key point: Many different random variables can be defined on the same sample space. Each asks a different question about the outcome.

Part 2. Probability Distributions

(How is probability distributed across possible values?)

From Random Variables to Distributions

A random variable tells us **what numerical values are possible**.

For our coin flip:

$$X = \begin{cases} 1 & \text{if Heads} \\ 0 & \text{if Tails} \end{cases}$$

But knowing the possible values is not enough.

How likely is each value?

For a fair coin:

$$P(X = 0) = 0.5 \quad P(X = 1) = 0.5$$

Probability Distribution

A **probability distribution** describes how the possible values of a random variable are distributed along with their chances of occurring.

Part 2-1. Discrete Random Variables and Probability Mass Function

Discrete Random Variables

Definition and Examples

Discrete Random Variable

A random variable X is **discrete** if its range (the set of values it can take) is **countable**—either finite or countably infinite.

Examples of discrete random variables:

- Coin flip: $X \in \{0, 1\}$ (*finite*)
- Die roll: $X \in \{1, 2, 3, 4, 5, 6\}$ (*finite*)
- Number of voters contacted before finding a Harris supporter (*countably infinite*)
- Number of children in a household (*finite, but the bound is large*)

Key feature: You can *list* all possible values, even if the list is infinitely long.

Probability Mass Function (PMF)

For a **discrete random variable**, the probability mass function (PMF) tells us the probability that X takes each possible value.

$$p(x) = P(X = x)$$

For a fair coin:

x	$p(x) = P(X = x)$
0	0.5
1	0.5

A valid PMF must satisfy:

$$p(x) \geq 0 \quad (\text{probabilities are non-negative})$$

$$\sum_x p(x) = 1 \quad (\text{probabilities sum to 1})$$

PMF: A Political Science Example

Let X = number of protests a respondent attended last year.

Suppose we have **100 respondents**:

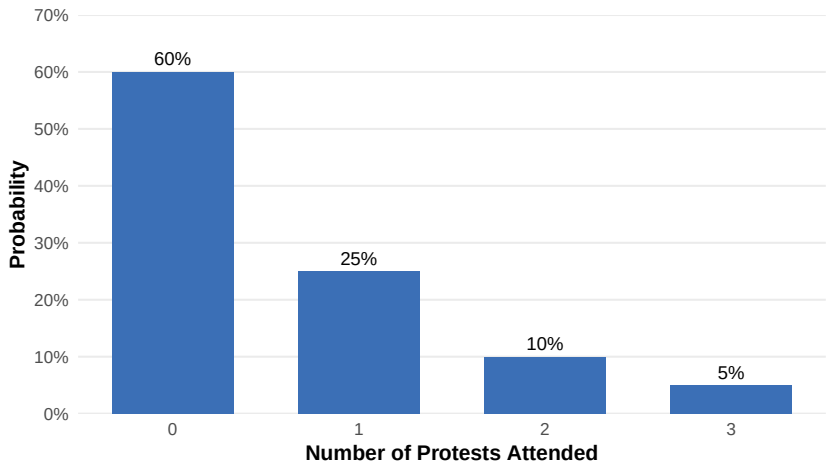
Protests Attended (x)	Number of Respondents	$P(X = x)$
0	60	$60/100 = 0.60$
1	25	$25/100 = 0.25$
2	10	$10/100 = 0.10$
3	5	$5/100 = 0.05$

For example,

$$P(X = 1) = \frac{25}{100} = 0.25$$

If we randomly select one respondent, there is a **25% probability** that they attended exactly one protest.

Probability Distribution of Protest Attendance



Part 2-2. Continuous Random Variables and Probability Density Function

Continuous Random Variables: The Problem

Consider X = height of a randomly selected adult (in inches).

Unlike a discrete random variable, X can take **any value** within an interval:

$$X \in [48, 84]$$

There are **uncountably many** possible values.

The Problem: What is the probability of one exact height?

$$P(X = 67.3842719\dots) = ?$$

Continuous Random Variables: The Problem

Problem: What is $P(X = 67.3842719\dots)$?

- There are infinitely many possible heights.
- If each had positive probability, they would sum to > 1 .
- So each specific value must have probability **zero**.

Consequence: For a continuous random variable,

$$P(X = x) = 0 \quad \text{for every } x$$

This seems paradoxical:

The probability of any specific outcome is zero, yet some outcome must occur.

Solution: Don't ask about specific values. Ask about **intervals**.

Probability Density Function (PDF)

For a **continuous random variable**, the probability density function (PDF) describes how probability is distributed across possible values.

$$f(x)$$

Unlike a PMF, the height $f(x)$ is **not itself a probability**.

Probability is given by the **area under the density curve**:

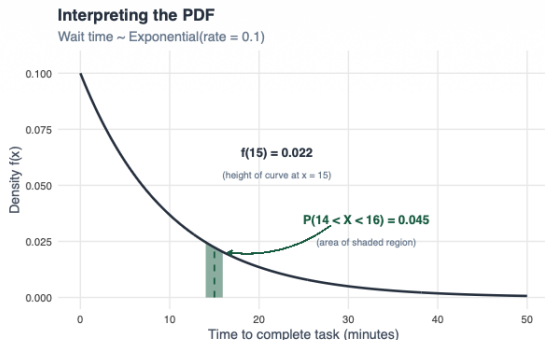
$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

A valid PDF must satisfy:

$$f(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

Interpreting the PDF

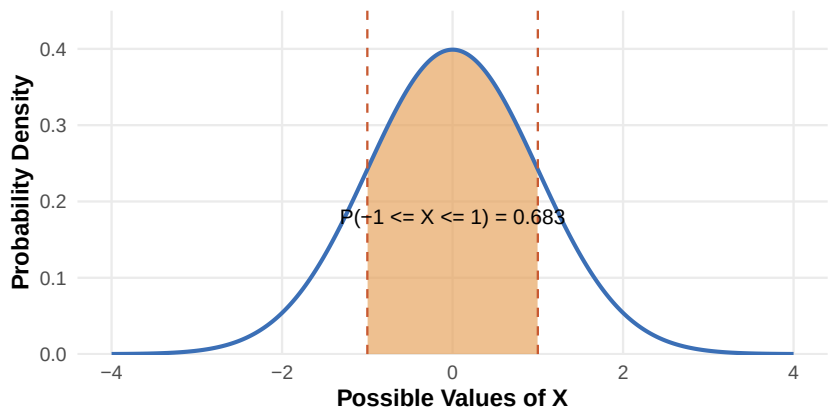
What does the height of the curve mean?



Plain English: The PDF tells you how “crowded” probability is near each value.

- **Height** $f(x)$ = probability per unit (not a probability itself)
- **Area** under curve = actual probability

Density of a Continuous Random Variable



$$P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \approx 0.683$$

X is a continuous random variable with a standard normal distribution, and the shaded area shows $P(-1 \leq X \leq 1) \approx 0.683$.

Part 2-3. Cumulative Distribution Function

From Probabilities to Cumulative Probabilities

So far, we have learned how to describe probabilities:

- **Discrete:** PMF gives probabilities at specific values

$$p(x) = P(X = x)$$

- **Continuous:** PDF gives probabilities over intervals

$$P(a \leq X \leq b)$$

But we often want to ask a different kind of question:

$$P(X \leq x)$$

*What is the probability that X is **less than or equal to** some value x ?*

Example: Protests

Protests Attended (x)	Number of Respondents	$P(X = x)$
0	60	$60/100 = 0.60$
1	25	$25/100 = 0.25$
2	10	$10/100 = 0.10$
3	5	$5/100 = 0.05$

For our protest example:

$$\begin{aligned}P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\&= 0.60 + 0.25 + 0.10 \\&= 0.95\end{aligned}$$

This is a **cumulative probability**.

Cumulative Distribution Function (CDF)

The **cumulative distribution function (CDF)** gives the probability that a random variable takes a value **less than or equal to** x .

$$F(x) = P(X \leq x)$$

The CDF is defined for **any random variable**:

- **Discrete** random variables
- **Continuous** random variables

Key Idea

$F(x)$ tells us how much probability has **accumulated** up to x .

CDF for a Discrete Random Variable

For a **discrete** random variable, we add up the probabilities of all possible values less than or equal to x :

$$F(x) = P(X \leq x) = \sum_{t \leq x} P(X = t)$$

Using our protest example:

x	$P(X = x)$	$F(x) = P(X \leq x)$
0	0.60	0.60
1	0.25	$0.60 + 0.25 = 0.85$
2	0.10	$0.60 + 0.25 + 0.10 = 0.95$
3	0.05	1.00

For example,

$$F(1) = P(X \leq 1) = P(X = 0) + P(X = 1) = 0.85$$

Interpretation: There is an **85% probability** that a randomly selected individual attended at most one protest.

CDF for a Continuous Random Variable

The same idea applies to a **continuous** random variable.

Suppose X = percentage of the vote received by a candidate.

The CDF still means:

$$F(x) = P(X \leq x)$$

For example,

$$F(50) = P(X \leq 50)$$

is the probability that the candidate receives **50% of the vote or less**.

For continuous random variables, probability accumulates over an **interval** rather than at individual values.

$$F(x) = \int_{-\infty}^x f(t) dt$$

Same Concept, Different Notation

Discrete: Add probability masses $\rightarrow \sum$

Continuous: Add probability density $\rightarrow \int$

Three Tools for Probability Distributions

PMF, PDF, and CDF

Each question has a corresponding tool:

Question	Tool	Works For
$P(X = x)$	PMF (probability mass function)	Discrete only
Density at x	PDF (probability density function)	Continuous only
$P(X \leq x)$	CDF (cumulative distribution function)	Both

Examples:

- Die roll: “ $P(\text{roll} = 4)?$ ” → Use **PMF**
- Temperature: “How dense is probability near 72° ?” → Use **PDF**
- Temperature: “ $P(\text{temp} \leq 70^\circ)?$ ” → Use **CDF**

Part 3. Expected Value

(The Center of Mass of a Distribution)

Summarizing a Probability Distribution

Why We Need Summary Statistics

A **probability distribution** gives us complete information about a random variable.

But sometimes we want a **single number** that summarizes the distribution:

- What's the average income in this population?
- What's the average approval rating?

The most common summary is the **expected value (or mean)**.

Intuition

If we could repeat the random process infinitely many times and average the results, what would we get?

Definition: Expected Value

Expected Value

The **expected value** (or **mean**) of a random variable X is:

$$\mathbb{E}[X] = \begin{cases} \sum x \cdot p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Interpretation: The expected value is a **weighted average** of possible values, where the weights are the probabilities (or densities).

Notation: $\mathbb{E}[X]$, $\bar{X}$, μ , and μ_X all denote the same thing.

The expected value need not be a possible value of X .

For example, the expected value of a fair die roll is 3.5, even though you can never actually roll 3.5.

Computing the Expected Value, Example

Let X = outcome of a fair die roll.

$$p(x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}$$

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=1}^6 x \cdot p(x) \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} \\ &= \frac{21}{6} = \mathbf{3.5}\end{aligned}$$

Interpretation: If you rolled a die many times and averaged the outcomes, you would get approximately **3.5**.

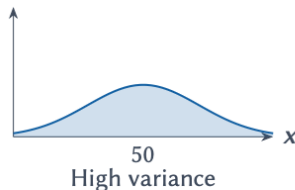
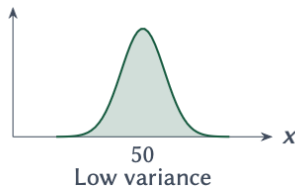
Part 4. Variance

(How spread out is the distribution)

Spread Matters

Why the mean isn't enough?

Consider two random variables, both with $\mathbb{E}[X] = 50$:



Both have the same mean, but they behave very differently.

Variance measures how spread out the distribution is around its mean.

Definition: Variance

Variance

The **variance** of a random variable X is:

$$\text{Var}(X) = \mathbb{E} [(X - \mathbb{E}[X])^2]$$

Interpretation: The average **squared deviation from the mean**.

Why squared?

- Deviations above and below the mean would **cancel out**
- Squaring makes all deviations **non-negative**
- Larger deviations get **penalized more heavily**

Notation: $\text{Var}(X)$, σ^2 , and σ_X^2 all denote the variance.

Computing the Variance, Example: Fair Die

Recall that X = outcome of a fair die roll:

$$p(x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}, \quad \mathbb{E}[X] = 3.5$$

Using

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2],$$

we get:

$$\begin{aligned} \text{Var}(X) &= \sum_{x=1}^6 (x - 3.5)^2 p(x) \\ &= \frac{1}{6} \left[(1 - 3.5)^2 + (2 - 3.5)^2 + (3 - 3.5)^2 \right. \\ &\quad \left. + (4 - 3.5)^2 + (5 - 3.5)^2 + (6 - 3.5)^2 \right] \\ &= \frac{17.5}{6} \\ &\approx \mathbf{2.92} \end{aligned}$$

Computing Variance Manually in R

Suppose we observe the number of protests attended by 6 respondents:

```
x <- c(0, 1, 0, 2, 3, 0)
```

Recall the definition:

$$\text{Var}(X) = \mathbb{E} [(X - \mathbb{E}[X])^2]$$

We can reproduce each step in R.

Step 1: Find the mean

```
x_bar <- mean(x)  
x_bar
```

```
## [1] 1
```

Computing Variance Manually in R

Step 2: Calculate deviations from the mean

```
deviation <- x - x_bar  
deviation
```

```
## [1] -1  0 -1  1  2 -1
```

Step 3: Square the deviations

```
squared_deviation <- deviation^2  
squared_deviation
```

```
## [1] 1 0 1 1 4 1
```

Computing Variance Manually in R

Step 4: Take their average

```
mean(squared_deviation)
```

```
## [1] 1.333333
```

Equivalently, we can write everything in one line:

```
mean((x - mean(x))^2)
```

```
## [1] 1.333333
```

Standard Deviation

Variance has one drawback: its units are **squared**.

If X is measured in dollars,

$\text{Var}(X)$ is measured in dollars².

This can be difficult to interpret.

Standard Deviation

The **standard deviation** of X is the square root of its variance:

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Interpretation: Standard deviation describes the **typical amount of variation around the mean**.

Unlike variance, standard deviation is measured in the **same units** as X .

From Variance to Covariance

So far, we have described **one random variable**:

- **Expected value:** Where is X centered?
- **Variance:** How much does X vary around its mean?
- **Standard deviation:** How much does X typically vary in its original units?

But social scientists are often interested in **relationships between two variables**:

- Does education increase with income?
- Does political knowledge increase with turnout?
- Does unemployment move with incumbent approval?

New Question

Do two random variables X and Y tend to **move together**?

This is what **covariance** measures.

Part 5. Covariance

(How do two variables move together?)

The Intuition Behind Covariance

Consider two random variables, X and Y .

For each observation, ask:

$$X - \mathbb{E}[X] \quad \text{and} \quad Y - \mathbb{E}[Y]$$

Are X and Y above or below their means **together**?

$X - \mathbb{E}[X]$	$Y - \mathbb{E}[Y]$	Product
+	+	+
-	-	+
+	-	-
-	+	-

Key Idea

Same direction $\rightarrow$ **positive covariance**

Opposite directions $\rightarrow$ **negative covariance**

Definition: Covariance

Covariance

The **covariance** between two random variables X and Y is:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

The formula calculates the average product of how far X and Y are from their respective means.

Interpretation:

- $\text{Cov}(X, Y) > 0$: X and Y tend to move in the **same direction**
- $\text{Cov}(X, Y) < 0$: X and Y tend to move in **opposite directions**
- $\text{Cov}(X, Y) \approx 0$: little **linear co-movement**

The Problem with Covariance

Covariance tells us whether two variables tend to move together:

$$\text{Cov}(X, Y) > 0 \quad \text{or} \quad \text{Cov}(X, Y) < 0$$

But its **magnitude depends on the units of measurement**.

Suppose income is measured in:

- **Dollars**
- **Thousands of dollars**

The covariance will change—even though the relationship is exactly the same.

The Problem

We need a measure of association that does **not depend on units**.

This leads us to **correlation**.

Why Covariance Depends on Units

Suppose we observe education and income for five individuals:

```
education <- c(12, 14, 16, 18, 20)
income_dollars <- c(30000, 40000, 50000, 65000, 80000)

# Calculate deviations from the means
dev_education <- education - mean(education)
dev_income <- income_dollars - mean(income_dollars)

# Multiply deviations and divide by n - 1
n <- length(education)
cov_dollars <- sum(dev_education * dev_income) / (n - 1)
cov_dollars

## [1] 62500

# Equivalently
cov(education, income_dollars)

## [1] 62500
```

Why Covariance Depends on Units

Now measure the **same income** in thousands of dollars:

```
income_thousands <- income_dollars / 1000
```

```
income_thousands
```

```
## [1] 30 40 50 65 80
```

```
dev_income_thousands <- income_thousands - mean(income_thousands)
```

```
cov_thousands <- sum(  
  dev_education * dev_income_thousands  
) / (n - 1)
```

```
cov_thousands
```

```
## [1] 62.5
```

```
# Equivalently  
cov(education, income_thousands)
```

```
## [1] 62.5
```

Definition: Correlation

Correlation

The **correlation** between X and Y is:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Correlation is simply **standardized covariance**.

Because we divide by the standard deviations of X and Y :

- Correlation has **no units**
- Correlation is always between -1 and 1

$$-1 \leq \rho_{XY} \leq 1$$

Correlation Does Not Depend on Units

Recall our education and income data:

```
education <- c(12, 14, 16, 18, 20)

income_dollars <- c(30000, 40000, 50000, 65000, 80000)

cor(education, income_dollars)
```

```
## [1] 0.9944466
```

Now measure income in **thousands of dollars**:

```
income_thousands <- income_dollars / 1000

cor(education, income_thousands)
```

```
## [1] 0.9944466
```

Key Take-away

Correlation is **unit-free**. Changing the scale of measurement does not change the strength of the relationship.

Interpreting Correlation

The sign tells us the **direction** of the relationship:

$$\rho > 0 \Rightarrow \text{Positive relationship}$$

$$\rho < 0 \Rightarrow \text{Negative relationship}$$

The absolute value tells us the **strength of the linear relationship**:

Correlation	Interpretation
$\rho = 1$	Perfect positive linear relationship
$\rho \approx 0$	Little or no linear relationship
$\rho = -1$	Perfect negative linear relationship

Important

Correlation measures **linear association**. $\rho = 0$ does not necessarily mean that X and Y are unrelated.

Key Takeaways

- **Random Variable:** Translates outcomes of a random process into numerical values.
- **Probability Distribution:** Describes the possible values of a random variable and how probability is distributed across them.
 - PMF: discrete random variables
 - PDF: continuous random variables
 - CDF: both discrete and continuous random variables
- **Expected Value:** Describes the **center** of a probability distribution.
- **Variance & Standard Deviation:** Describe how **spread out** a random variable is around its mean.
- **Covariance:** Describes whether two variables tend to **move together**.
- **Correlation:** Standardizes covariance to measure the **direction and strength of a linear relationship** on a scale from -1 to 1 .

The Big Picture

Random Variable → Distribution → Center & Spread → Relationships Between Variables